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OVER-THE-AIR CONFIGURATION AND MANAGEMENT FOR NEW RADIO-BASED SMART REPEATERS

Abstract

The described technology is generally directed to over-the-air configuration and management of new radio smart repeaters. To facilitate data forwarding, a smart repeater is configured to generate a forwarding layer that is logically above a physical layer of a mobile termination function (corresponding to a repeater link) and a physical layer of distributed unit function of the repeater equipment (corresponding to an access link). Via the forwarding layer, when traffic is received from a parent node, the traffic is forwarded via the access link to user equipment, and when traffic is received from a user equipment, the traffic is forwarded via the access link to a parent node. A smart repeater control management function/interface manages control plane signaling and configurations for the mobile termination function and distributed unit function.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/661,271 filed on Apr. 28, 2022. All sections of the aforementioned application are incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] The subject application is related to wireless communication systems, and, for example, to configuring and managing layer-1 (L1) based repeaters for advanced networks, including, but not limited to, fifth generation (5G) new radio (NR) networks and beyond, and related embodiments.

BACKGROUND

[0003] Due to the expected larger bandwidth available for new radio (NR) along with the native deployment of massive multiple input multiple output (MIMO) or multi-beam systems in NR, there is now an opportunity to develop and deploy smart repeaters (also called network controlled repeaters). A smart repeater node utilizes a layer-1 forwarding approach to take the downlink (DL) NR signal from a donor cell (e.g. macro distributed unit, or DU) received at a repeater link and provide the signal to user devices (user equipment, or UE) that may be beyond the coverage of the donor cell via a repeater access link. Additionally, the smart repeater can provide the uplink (UL) signal from the user device to its serving cell/donor DU. Because the signals on the access and repeater link are intended to be identical, the user can connect to the smart repeater in a transparent manner, (in contrast with an L2/L3 relay node such as Integrated Access and Backhaul (IAB)).

[0004] Conventional repeaters that operate using layer-1 forwarding approaches only attempt to replicate the repeater link received at the repeater node on an access link to the user devices. Because deployment of the repeater increases the signal strength of the parent node (donor DU), such deployment also may potentially increase interference to neighbor cells. Repeater transmissions also need to be carefully time-aligned with the transmissions from the donor DU in order avoid cross-link interference between DL and UL signals, particularly in the case of networks operating on time division duplex (TDD) bands. Furthermore, the utilization of dynamic analog and digital beamforming at gNodeBs (gNBs) and UE devices makes the deployment and management of conventional repeaters more challenging because the adaptation of the beamforming pattern and any sweeping of beams for control and access are dynamically managed over-the-air (OTA), and in some cases is user-specific (e.g. for channel status information (CSI) acquisition and data plane transmissions).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] Non-limiting and non-exhaustive embodiments of the subject disclosure are described with reference to the following figures, wherein like reference numerals refer to like parts throughout the various views unless otherwise specified.

[0006] FIG. 1 illustrates an example wireless communication system in which a repeater node communicates via a repeater link and access link, including communicating for over-the-air configuration and management, in accordance with various aspects and embodiments of the subject disclosure.

[0007] FIG. 2 is an example protocol stack of a smart repeater node including a smart repeater

control management function and interface, in accordance with various aspects and embodiments of the subject disclosure.

[0008] FIG. 3 illustrates an example of a smart repeater node configured for multiplexing, in accordance with various aspects and embodiments of the subject disclosure.

[0009] FIG. 4 illustrates an example of a smart repeater node obtaining configuration data, including synchronization signal blocks (SSBs) of the donor node to broadcast on the access link, in accordance with various aspects and embodiments of the subject disclosure, in accordance with various aspects and embodiments of the subject disclosure.

[0010] FIG. 5 illustrates an example of a smart repeater node obtaining configuration data, including configuration of signals/channels utilized as part of layer-1 data and control channels, in accordance with various aspects and embodiments of the subject disclosure.

[0011] FIG. 6 illustrates an example of a smart repeater node obtaining configuration data, including configuration of signals/channels utilized as part of layer-1 link adaptation, in accordance with various aspects and embodiments of the subject disclosure.

[0012] FIG. 7 is a flow diagram showing example operations related to facilitating the configuring of repeater equipment to generate a forwarding layer, in accordance with various aspects and embodiments of the subject disclosure.

[0013] FIG. 8 is a flow diagram showing example operations related to a repeater configuring a forwarding layer based on configuration data, in accordance with various aspects and embodiments of the subject disclosure.

[0014] FIG. 9 is a flow diagram showing example operations related to communicating via a configured repeater, in accordance with various aspects and embodiments of the subject disclosure.

[0015] FIG. 10 illustrates an example block diagram of an example user equipment that can be a mobile handset in accordance with various aspects and embodiments of the subject disclosure.

[0016] FIG. 11 illustrates an example block diagram of a computer that can be operable to execute processes and methods in accordance with various aspects and embodiments of the subject disclosure.

DETAILED DESCRIPTION

[0017] Various aspects of the technology described herein are directed towards over-the-air configuration and management of smart repeaters that can adapt the physical layer parameters used by the smart repeater for layer-1 forwarding operation. Described herein is configuring, managing and operating layer-1 based repeaters in 5G NR networks, including how to configure and manage air interface-related parameters using over-the-air signaling. In one implementation, the signaling can be based on already-defined Integrated Access and Backhaul (IAB) signaling protocols.

[0018] It should be understood that any of the examples and terms used herein are non-limiting. For instance, the examples are based on New Radio (NR, sometimes referred to as 5G) communications between a user equipment device exemplified as a smartphone, mobile device or the like and network node devices via a smart repeater; however virtually any communications devices may benefit from the technology described herein. Thus, any of the embodiments, aspects, concepts, structures, functionalities or examples described herein are non-limiting, and the technology may be used in various ways that provide benefits and advantages in radio communications in general.

[0019] In some embodiments the non-limiting term “radio network node” or simply “network node,” “radio network device or simply “network device” is used herein. These terms may be used interchangeably, and refer to any type of network node that serves user equipment and/or connected to other network node or network element or any radio node from where user equipment receives signal. Examples of radio network nodes are Node B, base station (BS), multi-standard radio (MSR) node such as MSR BS, gNodeB, eNode B, network controller, radio network controller (RNC), base station controller (BSC), relay, donor node controlling relay, base transceiver station (BTS), access point (AP), transmission points, transmission nodes, RRU, RRH, nodes in

distributed antenna system (DAS) etc.

[0020] In some embodiments the non-limiting term user equipment (UE) is used. It refers to any type of wireless device that communicates with a radio network node in a cellular or mobile communication system. Examples of user equipment are target device, device to device (D2D) user equipment, machine type user equipment or user equipment capable of machine to machine (M2M) communication, PDA, Tablet, mobile terminals, smart phone, laptop embedded equipped (LEE), laptop mounted equipment (LME), USB dongles etc.

[0021] Some embodiments are described in particular for 5G new radio systems. The embodiments are however applicable to any radio access technology (RAT) or multi-RAT system where the user equipment operates using multiple carriers e.g., LTE FDD/TDD, WCDMA/HSPA, GSM/GERAN, Wi Fi, WLAN, WiMax, CDMA2000 etc.

[0022] The embodiments are applicable to single carrier as well as to multicarrier (MC) or carrier aggregation (CA) operation of the user equipment. The term carrier aggregation (CA) is also called (e.g., interchangeably called) “multi-carrier system”, “multi-cell operation”, “multi-carrier operation”, “multi-carrier” transmission and/or reception. Note that the solutions outlined applies for Multi RAB (radio bearers) on some carriers (that is data plus speech is simultaneously scheduled).

[0023] In various embodiments, a network can be configured to provide and employ 5G wireless networking features and functionalities. With 5G networks that may use waveforms that split the bandwidth into several sub-bands, different types of services can be accommodated in different sub bands with the most suitable waveform and numerology, leading to improved spectrum utilization for 5G networks. Notwithstanding, in the mmWave spectrum, the millimeter waves have shorter wavelengths relative to other communications waves, whereby mmWave signals can experience severe path loss, penetration loss, and fading. However, the shorter wavelength at mmWave frequencies also allows more antennas to be packed in the same physical dimension, which allows for large-scale spatial multiplexing and highly directional beamforming.

[0024] Performance can be improved if both the transmitter and the receiver are equipped with multiple antennas. Multi-antenna techniques can significantly increase the data rates and reliability of a wireless communication system. The use of multiple input multiple output (MIMO) techniques, which was introduced in the third-generation partnership project (3GPP) and has been in use (including with LTE), is a multi-antenna technique that can improve the spectral efficiency of transmissions, thereby significantly boosting the overall data carrying capacity of wireless systems. The use of multiple-input multiple-output (MIMO) techniques can improve mmWave communications; MIMO can be used for achieving diversity gain, spatial multiplexing gain and beamforming gain.

[0025] Note that using multi-antennas does not always mean that MIMO is being used. For example, a configuration can have two downlink antennas, and these two antennas can be used in various ways. In addition to using the antennas in a 2×2 MIMO scheme, the two antennas can also be used in a diversity configuration rather than MIMO configuration. Even with multiple antennas, a particular scheme might only use one of the antennas (e.g., LTE specification's transmission mode 1, which uses a single transmission antenna and a single receive antenna). Or, only one antenna can be used, with various different multiplexing, precoding methods etc.

[0026] The MIMO technique uses a commonly known notation ($M \times N$) to represent MIMO configuration in terms number of transmit (M) and receive antennas (N) on one end of the transmission system. The common MIMO configurations used for various technologies are: (2×1) , (1×2) , (2×2) , (4×2) , (8×2) and (2×4) , (4×4) , (8×4) . The configurations represented by (2×1) and (1×2) are special cases of MIMO known as transmit diversity (or spatial diversity) and receive diversity. In addition to transmit diversity (or spatial diversity) and receive diversity, other techniques such as spatial multiplexing (comprising both open-loop and closed-loop), beamforming, and codebook-based precoding can also be used to address issues such as efficiency,

interference, and range.

[0027] FIG. 1 illustrates an example wireless communication system **100** comprising a parent node (macro/donor distributed unit (DU) **102**) communicating, via a smart repeater node **104**, with a user equipment (UE) **106**. In general, the smart repeater node **104** communicates with the donor DU **102** via a repeater link, and communicates with the UE **106** via an access link.

[0028] To act as repeater, the smart repeater node **104** is configured with a mobile termination (MT, or UE-like) function **108** comprising a layer-1 physical (PHY) layer corresponding to the repeater link. The smart repeater node **104** is also configured with a DU (gNB-like) function **110** comprising a layer-1 (L1) physical (PHY) layer corresponding to the access link. As also shown in FIG. 1 and as described herein, a smart repeater control (SR-C) management function acts as an SR-C interface **112** between the repeater's MT functionality and the repeater's DU functionality.

[0029] More particularly, FIG. 2 illustrates an exemplary protocol stack structure for the smart repeater node **104**. When the repeater links carrying the traffic from the donor node are based on the same channels and protocols as the access links carrying user data traffic, as shown in FIG. 2 the smart repeater node **104** can be constructed to contain two parallel protocol stacks. The mobile termination (MT) functionality provides connectivity between the smart repeater node and its parent donor node, which has a wired connection to the core network; (it is feasible for a repeater node to be a parent node to another repeater node and so on, however ultimately there is a donor DU node). The mobile termination (MT) functionality **220** thus includes a protocol stack comprising the PHY (MT function) layer **108**, medium access layer (MAC) layer **222** radio link control (RLC) layer **224** packet data convergence control (PDCP) layer **226** and radio resource control (RRC) layer **228**. The radio resource control **228** couples to the F1 Application Protocol (F1-AP) **230** of the distributed unit functionality **232**.

[0030] In general, the gNB function or distributed unit (DU) function **232** provides connectivity between the smart repeater node and access UEs. Note that on the DU function **232**, unlike IAB or other L2/L3 relay nodes, the smart repeater **104** only implements the physical layer of the DL and UL protocol stack because of the use of a layer-1 forwarding approach. This may facilitate easier deployment of a dense network of access points by building upon many of the control and data channels/procedures defined for providing access to UEs. An example is shown in FIG. 1, where the repeater node **104** can multiplex access and repeater links in time, frequency, or space (e.g. beam-based operation).

[0031] To this end, in one embodiment described herein and shown in FIG. 2, as part of the smart repeater air interface higher layer protocols, in order to route the repeater data plane traffic within the smart repeater node, an L1 forwarding layer **234** is generated and inserted above the PHY layer **108** of the MT function **220** and above the PHY layer **110** of the DU function **232** of the smart repeater node **104**. The L1 forwarding layer **234** facilitates the data forwarding operations of the smart repeater node **104**.

[0032] In addition to data forwarding, the smart repeater node **104** needs to manage the control plane signaling and configurations for both the MT and DU functions **220** and **232**, respectively. As shown in FIG. 2, an example of control plane signaling for the MT function involves layer-1 messages (e.g. physical downlink control channel (PDCCH) and/or Master Information Block (MIB)/system information (SIB) messages), medium access control (MAC) **222**, and radio resource control (RRC) **228**, along with the F1-AP interface **230** and operations, administration and maintenance (OAM) for the DU function **232**. This coordination is performed internally in the smart repeater node by the smart repeater control management function/(SR-C) interface **112**.

[0033] In one embodiment generally represented in FIG. 3, as part of the smart repeater air interface physical layer, the smart repeater nodes can multiplex the access and repeater links in time, frequency, and/or space (e.g. beam-based operation). Coordination of the multiplexing can be performed by the smart repeater control management function/(SR-C) interface **112**. Note that the same physical layer signals and channels used for these purposes by access UEs can be reused for

performing similar procedures at the smart repeater node **104**.

[0034] In another embodiment generally represented in FIG. **4**, the multiplexing at the smart repeater can include the configuration of signals/channels **440** utilized as part of layer 1 initial access and radio resource management. In one alternative, the smart repeater can be provided with a configuration of the synchronization signal blocks (SSBs) **442** of the donor node to broadcast on the access link. The configuration may be provided by physical layer or higher layer signaling and may comprise a SSB timing configuration (STC) **444** and/or a SSB measurement timing configuration (SMTTC) **446** that indicates the time and frequency resources (e.g. symbols and slots within a carrier bandwidth) for one or more SSBs and also their associated SSB indexes.

[0035] In a second alternative, a subset of the SSB configuration may be provided to the repeater node **104**, to cover only a subset of the time/frequency resources of the parent node, or to cover a reduced spatial (e.g. horizontal and/or elevation) domain compared to the parent node to mitigate coverage and interference considerations. In a third alternative, the smart repeater node **104** may be provided with a full or subset of time/frequency resources used for the random access channel (RACH) by the donor node. Such indication of RACH resources **448** may further comprise a set of random access preambles to be received by the smart repeater from access UEs during initial access or other connected mode procedures.

[0036] In another embodiment generally represented in FIG. **5**, the multiplexing at the smart repeater can include the configuration of signals/channels **550** utilized as part of layer 1 data and control channels. In one alternative the configuration may comprise an indication of supported frequency division duplex (FDD) bands **552** or time division duplex (TDD) bands **554**. In a second alternative, the configuration may comprise an indication of a supported bandwidth part (BWP) **556** including starting frequency location and size within the carrier bandwidth on an indication band. In a third alternative, the configuration may comprise an indication of a TDD downlink, uplink, and flexible slot pattern **558**. The indication of a TDD slot pattern may comprise a set of consecutive slots and may also indicate a set of not available (NA) slots where the smart repeater **104** does not transmit on the MT or DU function, and which may be instead reserved by the donor node for its own access link transmissions or measurements or may be used for measurement and reception of management messages at the smart repeater **104** itself.

[0037] In another embodiment generally represented in FIG. **6**, the multiplexing at the smart repeater includes the configuration of signals/channels **660** utilized as part of layer-1 link adaptation. In one alternative, the configuration comprises the configuration of resources **662** for the transmission of channel state information reference signals (CSI-RS) used by the donor node or a subset of CSI-RS resources. The configuration may indicate a set of CSI-RS indexes and a periodicity for transmission.

[0038] In a second alternative, the configuration **660** can include the configuration of resources **664** for the transmission of sounding reference signals (SRS) received by the donor node or a subset of SRS resources. The configuration may indicate a set of SRS patterns and a periodicity for reception. In a third alternative, the indication of beam data resources **668** used for link adaptation may include a set of analog and/or digital beams or beamforming weights which are to be used by the smart repeater on the access link and repeater link. In another alternative, the set of beams or beamforming weights may be different for the access link (smart repeater DU) and for the repeater link (smart repeater MT) in order to optimize system performance.

[0039] For any of the embodiments described herein, it should be noted that the physical layer or higher layer signaling used for configuration of the smart repeater may be scheduled (e.g., periodic) and configured by the parent node. In another alternative, the parent node may provide the indications in an aperiodic or “on-demand” manner, such as only when the parameters are modified. In another alternative, the smart repeater may request the parent node to provide a given resource configuration, and may request a set or subset of parameters and corresponding values based on a desired implementation at the smart repeater. These alternatives can be combined, e.g.,

periodic or other scheduled configuration can be planned, yet overridden on-demand when needed, and/or overridden by a smart repeater request. On-demand configuration can be overridden by a smart repeater request, and so on.

[0040] One or more aspects are represented in FIG. 7, and can comprise example operations, such as of a method. Example operation **702** represents facilitating, by network equipment comprising a processor, configuring repeater equipment to generate a forwarding layer that is logically above a physical layer of a mobile termination function of the repeater equipment and a distributed unit function of the repeater equipment, the configuring comprising configuring the repeater equipment to route at least data plane signals using the forwarding layer (operation **704**).

[0041] The configuring can further include configuring the repeater equipment to route control plane signals for the mobile termination function and the distributed unit function using the forwarding layer.

[0042] The configuring can further include configuring the repeater equipment to use the forwarding layer to multiplex, in time, signals across channels communicated via at least one: of an access link between the repeater equipment and a user equipment, or a repeater link between the repeater equipment and a parent node.

[0043] The configuring can further include configuring the repeater equipment to use the forwarding layer to multiplex, in frequency, signals across channels communicated via at least one of: an access link between the repeater equipment and a user equipment, or a repeater link between the repeater equipment and a parent node.

[0044] The configuring can further include configuring the repeater equipment to use the forwarding layer to multiplex, spatially, signals across channels communicated via at least one: of an access link between the repeater equipment and a user equipment, or a repeater link between the repeater equipment and a parent node.

[0045] The configuring can further include configuring the repeater equipment to use the forwarding layer to multiplex, in at least one of: time or frequency or spatially, signals across channels to be used as part of communications by the repeater equipment according to initial access and radio resource management protocol.

[0046] The configuring can further include configuring the repeater equipment to use parameters, of a configuration applicable to synchronization signal blocks of a donor node, to broadcast on an access link between the repeater equipment and a user equipment.

[0047] The configuration applicable to the synchronization signal blocks of the donor node can include at least one: of a synchronization signal block timing configuration, or a synchronization signal block measurement timing configuration. At least one of the synchronization signal block timing configuration or the synchronization signal block measurement timing configuration can indicate, to the repeater equipment, time and frequency resources applicable to the synchronization signal blocks and corresponding indexes for the synchronization signal blocks.

[0048] The configuring can further include configuring the repeater equipment to use a subgroup of parameters, of a group of parameters of a configuration applicable to synchronization signal blocks of a donor node, to broadcast on an access link between the repeater equipment and a user equipment, and wherein the subgroup of parameters indicates, to the repeater equipment, a subgroup of time and frequency resources applicable to the synchronization signal blocks and a corresponding subgroup of indexes for the synchronization signal blocks.

[0049] The configuring can further include configuring the repeater equipment to use at least a subgroup of parameters, of a group of parameters of a configuration, to indicate, to the repeater equipment, at least a subset of time and frequency resources applicable to communications on a random access channel between a donor node and the repeater equipment.

[0050] The configuring can further include configuring the repeater equipment to use a configuration applicable to data signals and control signals and channels utilized for the data signals and the control signals, and wherein the configuration indicates at least one of frequency

division duplex bands or time division duplex bands enabled for communication of the data signals and the control signals via the channels.

[0051] The configuring the repeater equipment further can include configuring the repeater equipment to use a configuration, the configuration comprising at least one of: a supported bandwidth part comprising starting frequency location data and size data within a carrier bandwidth on an indication band, an indication of a time division duplex information for downlink signals and uplink signals, a slot pattern comprising a set of consecutive slots, a slot pattern comprising a set of consecutive slots and a set of unavailable slots, signal data and channel data utilized for the signal data, channel state information reference signal data, channel state information reference signal index data, transmission timing data, sounding reference signal resource data, sounding reference signal pattern data, reception timing data, an indication of link adaptation resource data, analog beam data, digital beam data, beamforming weight data for use by the smart repeater on an access link between the repeater equipment and a user equipment, or beamforming weight data for use by the smart repeater on a repeater link between the repeater equipment and a parent node.

[0052] One or more aspects are represented in FIG. 8, and can comprise example operations, such as of a processor and a memory that stores executable instructions or components, that, when executed by the processor, facilitate performance of the example operations. Example operation **802** represents receiving, by a mobile termination function of repeater equipment, configuration data from a parent node. Example operation **804** represents, in response to the receiving, configuring a forwarding layer logically above a first physical layer of the mobile termination function and a second physical layer of a distributed unit function of the repeater equipment to forward data plane signals received at the first physical layer via the second physical layer to a user equipment.

[0053] Further operations can include managing, by the repeater equipment based on the configuration data, the control plane signaling and configuration for the mobile termination function, and the control plane signaling and configuration for the distributed unit function.

[0054] Further operations can include multiplexing, by the repeater equipment based on the configuration data, repeater link data communicated via the first physical layer and access link data communicated via the second physical layer.

[0055] Receiving the configuration data can include at least one of: receiving the configuration data from the parent node based on a schedule, receiving the configuration data from the parent node based on modified parameter data, or receiving the configuration data from the parent node based on a request from the repeater equipment.

[0056] One or more aspects are represented in FIG. 9, and can comprise a machine-readable medium, comprising executable instructions that, when executed by a processor, facilitate performance of example operations. Example operation **902** represents receiving, at a repeater link corresponding to a first physical layer of a mobile termination function of repeater equipment, data plane traffic from a parent node. Example operation **904** represents forwarding, via a forwarding layer configured logically above the first physical layer and a second physical layer of a distributed unit function of the repeater equipment, the data plane traffic to an access link corresponding to the second physical layer. Example operation **906** represents communicating the data plane traffic, via the access link, to a user equipment.

[0057] The data plane traffic can include first data plane traffic, and further operations can include receiving, via the access link, second data plane traffic from the user equipment, and forwarding, via the forwarding layer, the second data plane traffic to the first physical layer for communication of the second data plane traffic to the parent node via the repeater link.

[0058] Further operations can include multiplexing first communications of the repeater link and second communications of the access link.

[0059] As can be seen, the technology described herein facilitates over-the-air configuration and coordination of time, frequency, and spatial (e.g. beam-based) resources for NR-based smart

repeaters. The technology described herein facilitates, for a given smart repeater, indication of initial access and radio resource management, indication of data and control channel resources and/or indication of link adaptation resources, using, for example, signaling defined for IAB-MT functionality.

[0060] Turning to aspects in general, a wireless communication system can employ various cellular systems, technologies, and modulation schemes to facilitate wireless radio communications between devices (e.g., a UE and the network equipment). While example embodiments might be described for 5G new radio (NR) systems, the embodiments can be applicable to any radio access technology (RAT) or multi-RAT system where the UE operates using multiple carriers e.g., LTE FDD/TDD, GSM/GERAN, CDMA2000 etc. For example, the system can operate in accordance with global system for mobile communications (GSM), universal mobile telecommunications service (UMTS), long term evolution (LTE), LTE frequency division duplexing (LTE FDD, LTE time division duplexing (TDD), high speed packet access (HSPA), code division multiple access (CDMA), wideband CDMA (WCDMA), CDMA2000, time division multiple access (TDMA), frequency division multiple access (FDMA), multi-carrier code division multiple access (MC-CDMA), single-carrier code division multiple access (SC-CDMA), single-carrier FDMA (SC-FDMA), orthogonal frequency division multiplexing (OFDM), discrete Fourier transform spread OFDM (DFT-spread OFDM) single carrier FDMA (SC-FDMA), Filter bank based multi-carrier (FBMC), zero tail DFT-spread-OFDM (ZT DFT-s-OFDM), generalized frequency division multiplexing (GFDM), fixed mobile convergence (FMC), universal fixed mobile convergence (UFMC), unique word OFDM (UW-OFDM), unique word DFT-spread OFDM (UW DFT-Spread-OFDM), cyclic prefix OFDM CP-OFDM, resource-block-filtered OFDM, Wi Fi, WLAN, WiMax, and the like. However, various features and functionalities of system are particularly described wherein the devices (e.g., the UEs and the network equipment) of the system are configured to communicate wireless signals using one or more multi carrier modulation schemes, wherein data symbols can be transmitted simultaneously over multiple frequency subcarriers (e.g., OFDM, CP-OFDM, DFT-spread OFDM, UFMC, FBMC, etc.). The embodiments are applicable to single carrier as well as to multicarrier (MC) or carrier aggregation (CA) operation of the UE. The term carrier aggregation (CA) is also called (e.g., interchangeably called) “multi-carrier system”, “multi-cell operation”, “multi-carrier operation”, “multi-carrier” transmission and/or reception. Note that some embodiments are also applicable for Multi RAB (radio bearers) on some carriers (that is data plus speech is simultaneously scheduled).

[0061] In various embodiments, the system can be configured to provide and employ 5G wireless networking features and functionalities. With 5G networks that may use waveforms that split the bandwidth into several sub-bands, different types of services can be accommodated in different sub-bands with the most suitable waveform and numerology, leading to improved spectrum utilization for 5G networks. Notwithstanding, in the mmWave spectrum, the millimeter waves have shorter wavelengths relative to other communications waves, whereby mmWave signals can experience severe path loss, penetration loss, and fading. However, the shorter wavelength at mmWave frequencies also allows more antennas to be packed in the same physical dimension, which allows for large-scale spatial multiplexing and highly directional beamforming.

[0062] Performance can be improved if both the transmitter and the receiver are equipped with multiple antennas. Multi-antenna techniques can significantly increase the data rates and reliability of a wireless communication system. The use of multiple input multiple output (MIMO) techniques, which was introduced in the third-generation partnership project (3GPP) and has been in use (including with LTE), is a multi-antenna technique that can improve the spectral efficiency of transmissions, thereby significantly boosting the overall data carrying capacity of wireless systems. The use of multiple-input multiple-output (MIMO) techniques can improve mmWave communications; MIMO can be used for achieving diversity gain, spatial multiplexing gain and beamforming gain.

[0063] Note that using multi-antennas does not always mean that MIMO is being used. For example, a configuration can have two downlink antennas, and these two antennas can be used in various ways. In addition to using the antennas in a 2×2 MIMO scheme, the two antennas can also be used in a diversity configuration rather than MIMO configuration. Even with multiple antennas, a particular scheme might only use one of the antennas (e.g., LTE specification's transmission mode 1, which uses a single transmission antenna and a single receive antenna). Or, only one antenna can be used, with various different multiplexing, precoding methods etc.

[0064] The MIMO technique uses a commonly known notation (M×N) to represent MIMO configuration in terms number of transmit (M) and receive antennas (N) on one end of the transmission system. The common MIMO configurations used for various technologies are: (2×1), (1×2), (2×2), (4×2), (8×2) and (2×4), (4×4), (8×4). The configurations represented by (2×1) and (1×2) are special cases of MIMO known as transmit diversity (or spatial diversity) and receive diversity. In addition to transmit diversity (or spatial diversity) and receive diversity, other techniques such as spatial multiplexing (comprising both open-loop and closed-loop), beamforming, and codebook-based precoding can also be used to address issues such as efficiency, interference, and range.

[0065] Referring now to FIG. 10, illustrated is a schematic block diagram of an example end-user device such as a user equipment) that can be a mobile device **1000** capable of connecting to a network in accordance with some embodiments described herein. Although a mobile handset **1000** is illustrated herein, it will be understood that other devices can be a mobile device, and that the mobile handset **1000** is merely illustrated to provide context for the embodiments of the various embodiments described herein. The following discussion is intended to provide a brief, general description of an example of a suitable environment **1000** in which the various embodiments can be implemented. While the description includes a general context of computer-executable instructions embodied on a machine-readable storage medium, those skilled in the art will recognize that the various embodiments also can be implemented in combination with other program modules and/or as a combination of hardware and software.

[0066] Generally, applications (e.g., program modules) can include routines, programs, components, data structures, etc., that perform particular tasks or implement particular abstract data types. Moreover, those skilled in the art will appreciate that the methods described herein can be practiced with other system configurations, including single-processor or multiprocessor systems, minicomputers, mainframe computers, as well as personal computers, hand-held computing devices, microprocessor-based or programmable consumer electronics, and the like, each of which can be operatively coupled to one or more associated devices.

[0067] A computing device can typically include a variety of machine-readable media. Machine-readable media can be any available media that can be accessed by the computer and includes both volatile and non-volatile media, removable and non-removable media. By way of example and not limitation, computer-readable media can include computer storage media and communication media. Computer storage media can include volatile and/or non-volatile media, removable and/or non-removable media implemented in any method or technology for storage of information, such as computer-readable instructions, data structures, program modules or other data. Computer storage media can include, but is not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD ROM, digital video disk (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by the computer.

[0068] Communication media typically embodies computer-readable instructions, data structures, program modules or other data in a modulated data signal such as a carrier wave or other transport mechanism, and includes any information delivery media. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a manner as to encode

information in the signal. By way of example, and not limitation, communication media includes wired media such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media. Combinations of the any of the above should also be included within the scope of computer-readable media.

[0069] The handset **1000** includes a processor **1002** for controlling and processing all onboard operations and functions. A memory **1004** interfaces to the processor **1002** for storage of data and one or more applications **1006** (e.g., a video player software, user feedback component software, etc.). Other applications can include voice recognition of predetermined voice commands that facilitate initiation of the user feedback signals. The applications **1006** can be stored in the memory **1004** and/or in a firmware **1008**, and executed by the processor **1002** from either or both the memory **1004** or/and the firmware **1008**. The firmware **1008** can also store startup code for execution in initializing the handset **1000**. A communications component **1010** interfaces to the processor **1002** to facilitate wired/wireless communication with external systems, e.g., cellular networks, VoIP networks, and so on. Here, the communications component **1010** can also include a suitable cellular transceiver **1011** (e.g., a GSM transceiver) and/or an unlicensed transceiver **1013** (e.g., Wi-Fi, WiMax) for corresponding signal communications. The handset **1000** can be a device such as a cellular telephone, a PDA with mobile communications capabilities, and messaging-centric devices. The communications component **1010** also facilitates communications reception from terrestrial radio networks (e.g., broadcast), digital satellite radio networks, and Internet-based radio services networks.

[0070] The handset **1000** includes a display **1012** for displaying text, images, video, telephony functions (e.g., a Caller ID function), setup functions, and for user input. For example, the display **1012** can also be referred to as a “screen” that can accommodate the presentation of multimedia content (e.g., music metadata, messages, wallpaper, graphics, etc.). The display **1012** can also display videos and can facilitate the generation, editing and sharing of video quotes. A serial I/O interface **1014** is provided in communication with the processor **1002** to facilitate wired and/or wireless serial communications (e.g., USB, and/or IEEE 1094) through a hardwire connection, and other serial input devices (e.g., a keyboard, keypad, and mouse). This supports updating and troubleshooting the handset **1000**, for example. Audio capabilities are provided with an audio I/O component **1016**, which can include a speaker for the output of audio signals related to, for example, indication that the user pressed the proper key or key combination to initiate the user feedback signal. The audio I/O component **1016** also facilitates the input of audio signals through a microphone to record data and/or telephony voice data, and for inputting voice signals for telephone conversations.

[0071] The handset **1000** can include a slot interface **1018** for accommodating a SIC (Subscriber Identity Component) in the form factor of a card Subscriber Identity Module (SIM) or universal SIM **1020**, and interfacing the SIM card **1020** with the processor **1002**. However, it is to be appreciated that the SIM card **1020** can be manufactured into the handset **1000**, and updated by downloading data and software.

[0072] The handset **1000** can process IP data traffic through the communication component **1010** to accommodate IP traffic from an IP network such as, for example, the Internet, a corporate intranet, a home network, a person area network, etc., through an ISP or broadband cable provider. Thus, VoIP traffic can be utilized by the handset **800** and IP-based multimedia content can be received in either an encoded or decoded format.

[0073] A video processing component **1022** (e.g., a camera) can be provided for decoding encoded multimedia content. The video processing component **1022** can aid in facilitating the generation, editing and sharing of video quotes. The handset **1000** also includes a power source **1024** in the form of batteries and/or an AC power subsystem, which power source **1024** can interface to an external power system or charging equipment (not shown) by a power I/O component **1026**.

[0074] The handset **1000** can also include a video component **1030** for processing video content

received and, for recording and transmitting video content. For example, the video component **1030** can facilitate the generation, editing and sharing of video quotes. A location tracking component **1032** facilitates geographically locating the handset **1000**. As described hereinabove, this can occur when the user initiates the feedback signal automatically or manually. A user input component **1034** facilitates the user initiating the quality feedback signal. The user input component **1034** can also facilitate the generation, editing and sharing of video quotes. The user input component **1034** can include such conventional input device technologies such as a keypad, keyboard, mouse, stylus pen, and/or touch screen, for example.

[0075] Referring again to the applications **1006**, a hysteresis component **1036** facilitates the analysis and processing of hysteresis data, which is utilized to determine when to associate with the access point. A software trigger component **1038** can be provided that facilitates triggering of the hysteresis component **1038** when the Wi-Fi transceiver **1013** detects the beacon of the access point. A SIP client **1040** enables the handset **1000** to support SIP protocols and register the subscriber with the SIP registrar server. The applications **1006** can also include a client **1042** that provides at least the capability of discovery, play and store of multimedia content, for example, music.

[0076] The handset **1000**, as indicated above related to the communications component **810**, includes an indoor network radio transceiver **1013** (e.g., Wi-Fi transceiver). This function supports the indoor radio link, such as IEEE 802.11, for the dual-mode GSM handset **1000**. The handset **1000** can accommodate at least satellite radio services through a handset that can combine wireless voice and digital radio chipsets into a single handheld device.

[0077] In order to provide additional context for various embodiments described herein, FIG. **11** and the following discussion are intended to provide a brief, general description of a suitable computing environment **1100** in which the various embodiments of the embodiment described herein can be implemented. While the embodiments have been described above in the general context of computer-executable instructions that can run on one or more computers, those skilled in the art will recognize that the embodiments can be also implemented in combination with other program modules and/or as a combination of hardware and software.

[0078] Generally, program modules include routines, programs, components, data structures, etc., that perform particular tasks or implement particular abstract data types. Moreover, those skilled in the art will appreciate that the various methods can be practiced with other computer system configurations, including single-processor or multiprocessor computer systems, minicomputers, mainframe computers, Internet of Things (IoT) devices, distributed computing systems, as well as personal computers, hand-held computing devices, microprocessor-based or programmable consumer electronics, and the like, each of which can be operatively coupled to one or more associated devices.

[0079] The illustrated embodiments of the embodiments herein can be also practiced in distributed computing environments where certain tasks are performed by remote processing devices that are linked through a communications network. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

[0080] Computing devices typically include a variety of media, which can include computer-readable storage media, machine-readable storage media, and/or communications media, which two terms are used herein differently from one another as follows. Computer-readable storage media or machine-readable storage media can be any available storage media that can be accessed by the computer and includes both volatile and nonvolatile media, removable and non-removable media. By way of example, and not limitation, computer-readable storage media or machine-readable storage media can be implemented in connection with any method or technology for storage of information such as computer-readable or machine-readable instructions, program modules, structured data or unstructured data.

[0081] Computer-readable storage media can include, but are not limited to, random access memory (RAM), read only memory (ROM), electrically erasable programmable read only memory

(EEPROM), flash memory or other memory technology, compact disk read only memory (CD-ROM), digital versatile disk (DVD), Blu-ray disc (BD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, solid state drives or other solid state storage devices, or other tangible and/or non-transitory media which can be used to store desired information. In this regard, the terms “tangible” or “non-transitory” herein as applied to storage, memory or computer-readable media, are to be understood to exclude only propagating transitory signals per se as modifiers and do not relinquish rights to all standard storage, memory or computer-readable media that are not only propagating transitory signals per se.

[0082] Computer-readable storage media can be accessed by one or more local or remote computing devices, e.g., via access requests, queries or other data retrieval protocols, for a variety of operations with respect to the information stored by the medium.

[0083] Communications media typically embody computer-readable instructions, data structures, program modules or other structured or unstructured data in a data signal such as a modulated data signal, e.g., a carrier wave or other transport mechanism, and includes any information delivery or transport media. The term “modulated data signal” or signals refers to a signal that has one or more of its characteristics set or changed in such a manner as to encode information in one or more signals. By way of example, and not limitation, communication media include wired media, such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media.

[0084] With reference again to FIG. 11, the example environment **1100** for implementing various embodiments of the aspects described herein includes a computer **1102**, the computer **1102** including a processing unit **1104**, a system memory **1106** and a system bus **1108**. The system bus **1108** couples system components including, but not limited to, the system memory **1106** to the processing unit **1104**. The processing unit **1104** can be any of various commercially available processors. Dual microprocessors and other multi-processor architectures can also be employed as the processing unit **1104**.

[0085] The system bus **1108** can be any of several types of bus structure that can further interconnect to a memory bus (with or without a memory controller), a peripheral bus, and a local bus using any of a variety of commercially available bus architectures. The system memory **1106** includes ROM **1110** and RAM **1112**. A basic input/output system (BIOS) can be stored in a non-volatile memory such as ROM, erasable programmable read only memory (EPROM), EEPROM, which BIOS contains the basic routines that help to transfer information between elements within the computer **1102**, such as during startup. The RAM **1112** can also include a high-speed RAM such as static RAM for caching data.

[0086] The computer **1102** further includes an internal hard disk drive (HDD) **1114** (e.g., EIDE, SATA), one or more external storage devices **1116** (e.g., a magnetic floppy disk drive (FDD) **1116**, a memory stick or flash drive reader, a memory card reader, etc.) and an optical disk drive **1120** (e.g., which can read or write from a CD-ROM disc, a DVD, a BD, etc.). While the internal HDD **1114** is illustrated as located within the computer **1102**, the internal HDD **1114** can also be configured for external use in a suitable chassis (not shown). Additionally, while not shown in environment **1100**, a solid state drive (SSD), non-volatile memory and other storage technology could be used in addition to, or in place of, an HDD **1114**, and can be internal or external. The HDD **1114**, external storage device(s) **1116** and optical disk drive **1120** can be connected to the system bus **1108** by an HDD interface **1124**, an external storage interface **1126** and an optical drive interface **1128**, respectively. The interface **1124** for external drive implementations can include at least one or both of Universal Serial Bus (USB) and Institute of Electrical and Electronics Engineers (IEEE) 1094 interface technologies. Other external drive connection technologies are within contemplation of the embodiments described herein.

[0087] The drives and their associated computer-readable storage media provide nonvolatile

storage of data, data structures, computer-executable instructions, and so forth. For the computer **1102**, the drives and storage media accommodate the storage of any data in a suitable digital format. Although the description of computer-readable storage media above refers to respective types of storage devices, it should be appreciated by those skilled in the art that other types of storage media which are readable by a computer, whether presently existing or developed in the future, could also be used in the example operating environment, and further, that any such storage media can contain computer-executable instructions for performing the methods described herein.

[0088] A number of program modules can be stored in the drives and RAM **1112**, including an operating system **1130**, one or more application programs **1132**, other program modules **1134** and program data **1136**. All or portions of the operating system, applications, modules, and/or data can also be cached in the RAM **1112**. The systems and methods described herein can be implemented utilizing various commercially available operating systems or combinations of operating systems.

[0089] Computer **1102** can optionally include emulation technologies. For example, a hypervisor (not shown) or other intermediary can emulate a hardware environment for operating system **1130**, and the emulated hardware can optionally be different from the hardware illustrated in FIG. **11**. In such an embodiment, operating system **1130** can include one virtual machine (VM) of multiple VMs hosted at computer **1102**. Furthermore, operating system **1130** can provide runtime environments, such as the Java runtime environment or the .NET framework, for applications **1132**. Runtime environments are consistent execution environments that allow applications **1132** to run on any operating system that includes the runtime environment. Similarly, operating system **1130** can support containers, and applications **1132** can be in the form of containers, which are lightweight, standalone, executable packages of software that include, e.g., code, runtime, system tools, system libraries and settings for an application.

[0090] Further, computer **1102** can be enabled with a security module, such as a trusted processing module (TPM). For instance with a TPM, boot components hash next in time boot components, and wait for a match of results to secured values, before loading a next boot component. This process can take place at any layer in the code execution stack of computer **1102**, e.g., applied at the application execution level or at the operating system (OS) kernel level, thereby enabling security at any level of code execution.

[0091] A user can enter commands and information into the computer **1102** through one or more wired/wireless input devices, e.g., a keyboard **1138**, a touch screen **1140**, and a pointing device, such as a mouse **1142**. Other input devices (not shown) can include a microphone, an infrared (IR) remote control, a radio frequency (RF) remote control, or other remote control, a joystick, a virtual reality controller and/or virtual reality headset, a game pad, a stylus pen, an image input device, e.g., camera(s), a gesture sensor input device, a vision movement sensor input device, an emotion or facial detection device, a biometric input device, e.g., fingerprint or iris scanner, or the like. These and other input devices are often connected to the processing unit **1104** through an input device interface **1144** that can be coupled to the system bus **1108**, but can be connected by other interfaces, such as a parallel port, an IEEE 1094 serial port, a game port, a USB port, an IR interface, a BLUETOOTH® interface, etc.

[0092] A monitor **1146** or other type of display device can be also connected to the system bus **1108** via an interface, such as a video adapter **1148**. In addition to the monitor **1146**, a computer typically includes other peripheral output devices (not shown), such as speakers, printers, etc.

[0093] The computer **1102** can operate in a networked environment using logical connections via wired and/or wireless communications to one or more remote computers, such as a remote computer(s) **1150**. The remote computer(s) **1150** can be a workstation, a server computer, a router, a personal computer, portable computer, microprocessor-based entertainment appliance, a peer device or other common network node, and typically includes many or all of the elements described relative to the computer **1102**, although, for purposes of brevity, only a memory/storage device **1152** is illustrated. The logical connections depicted include wired/wireless connectivity to a

local area network (LAN) **1154** and/or larger networks, e.g., a wide area network (WAN) **1156**. Such LAN and WAN networking environments are commonplace in offices and companies, and facilitate enterprise-wide computer networks, such as intranets, all of which can connect to a global communications network, e.g., the Internet.

[0094] When used in a LAN networking environment, the computer **1102** can be connected to the local network **1154** through a wired and/or wireless communication network interface or adapter **1158**. The adapter **1158** can facilitate wired or wireless communication to the LAN **1154**, which can also include a wireless access point (AP) disposed thereon for communicating with the adapter **1158** in a wireless mode.

[0095] When used in a WAN networking environment, the computer **1102** can include a modem **1160** or can be connected to a communications server on the WAN **1156** via other means for establishing communications over the WAN **1156**, such as by way of the Internet. The modem **1160**, which can be internal or external and a wired or wireless device, can be connected to the system bus **1108** via the input device interface **1144**. In a networked environment, program modules depicted relative to the computer **1102** or portions thereof, can be stored in the remote memory/storage device **1152**. It will be appreciated that the network connections shown are example and other means of establishing a communications link between the computers can be used.

[0096] When used in either a LAN or WAN networking environment, the computer **1102** can access cloud storage systems or other network-based storage systems in addition to, or in place of, external storage devices **1116** as described above. Generally, a connection between the computer **1102** and a cloud storage system can be established over a LAN **1154** or WAN **1156** e.g., by the adapter **1158** or modem **1160**, respectively. Upon connecting the computer **1102** to an associated cloud storage system, the external storage interface **1126** can, with the aid of the adapter **1158** and/or modem **1160**, manage storage provided by the cloud storage system as it would other types of external storage. For instance, the external storage interface **1126** can be configured to provide access to cloud storage sources as if those sources were physically connected to the computer **1102**.

[0097] The computer **1102** can be operable to communicate with any wireless devices or entities operatively disposed in wireless communication, e.g., a printer, scanner, desktop and/or portable computer, portable data assistant, communications satellite, any piece of equipment or location associated with a wirelessly detectable tag (e.g., a kiosk, news stand, store shelf, etc.), and telephone. This can include Wireless Fidelity (Wi-Fi) and BLUETOOTH® wireless technologies. Thus, the communication can be a predefined structure as with a conventional network or simply an ad hoc communication between at least two devices.

[0098] The computer is operable to communicate with any wireless devices or entities operatively disposed in wireless communication, e.g., a printer, scanner, desktop and/or portable computer, portable data assistant, communications satellite, any piece of equipment or location associated with a wirelessly detectable tag (e.g., a kiosk, news stand, restroom), and telephone. This includes at least Wi-Fi and Bluetooth™ wireless technologies. Thus, the communication can be a predefined structure as with a conventional network or simply an ad hoc communication between at least two devices.

[0099] Wi-Fi, or Wireless Fidelity, allows connection to the Internet from a couch at home, a bed in a hotel room, or a conference room at work, without wires. Wi-Fi is a wireless technology similar to that used in a cell phone that enables such devices, e.g., computers, to send and receive data indoors and out; anywhere within the range of a base station. Wi-Fi networks use radio technologies called IEEE802.11 (a, b, g, n, etc.) to provide secure, reliable, fast wireless connectivity. A Wi-Fi network can be used to connect computers to each other, to the Internet, and to wired networks (which use IEEE802.3 or Ethernet). Wi-Fi networks operate in the unlicensed 2.4 and 8 GHz radio bands, at an 11 Mbps (802.11b) or 84 Mbps (802.11a) data rate, for example, or with products that contain both bands (dual band), so the networks can provide real-world

performance similar to the basic “10BaseT” wired Ethernet networks used in many offices.

[0100] As it employed in the subject specification, the term “processor” can refer to substantially any computing processing unit or device comprising, but not limited to comprising, single-core processors; single-processors with software multithread execution capability; multi-core processors; multi-core processors with software multithread execution capability; multi-core processors with hardware multithread technology; parallel platforms; and parallel platforms with distributed shared memory. Additionally, a processor can refer to an integrated circuit, an application specific integrated circuit (ASIC), a digital signal processor (DSP), a field programmable gate array (FPGA), a programmable logic controller (PLC), a complex programmable logic device (CPLD), a discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. Processors can exploit nano-scale architectures such as, but not limited to, molecular and quantum-dot based transistors, switches and gates, in order to optimize space usage or enhance performance of user equipment. A processor also can be implemented as a combination of computing processing units.

[0101] In the subject specification, terms such as “store,” “data store,” “data storage,” “database,” “repository,” “queue”, and substantially any other information storage component relevant to operation and functionality of a component, refer to “memory components,” or entities embodied in a “memory” or components comprising the memory. It will be appreciated that the memory components described herein can be either volatile memory or nonvolatile memory, or can include both volatile and nonvolatile memory. In addition, memory components or memory elements can be removable or stationary. Moreover, memory can be internal or external to a device or component, or removable or stationary. Memory can include various types of media that are readable by a computer, such as hard-disc drives, zip drives, magnetic cassettes, flash memory cards or other types of memory cards, cartridges, or the like.

[0102] By way of illustration, and not limitation, nonvolatile memory can include read only memory (ROM), programmable ROM (PROM), electrically programmable ROM (EPROM), electrically erasable ROM (EEPROM), or flash memory. Volatile memory can include random access memory (RAM), which acts as external cache memory. By way of illustration and not limitation, RAM is available in many forms such as synchronous RAM (SRAM), dynamic RAM (DRAM), synchronous DRAM (SDRAM), double data rate SDRAM (DDR SDRAM), enhanced SDRAM (ESDRAM), Synchlink DRAM (SLDRAM), and direct Rambus RAM (DRRAM). Additionally, the disclosed memory components of systems or methods herein are intended to include, without being limited, these and any other suitable types of memory.

[0103] In particular and in regard to the various functions performed by the above described components, devices, circuits, systems and the like, the terms (including a reference to a “means”) used to describe such components are intended to correspond, unless otherwise indicated, to any component which performs the specified function of the described component (e.g., a functional equivalent), even though not structurally equivalent to the disclosed structure, which performs the function in the herein illustrated example aspects of the embodiments. In this regard, it will also be recognized that the embodiments include a system as well as a computer-readable medium having computer-executable instructions for performing the acts and/or events of the various methods.

[0104] Computing devices typically include a variety of media, which can include computer-readable storage media and/or communications media, which two terms are used herein differently from one another as follows. Computer-readable storage media can be any available storage media that can be accessed by the computer and includes both volatile and nonvolatile media, removable and non-removable media. By way of example, and not limitation, computer-readable storage media can be implemented in connection with any method or technology for storage of information such as computer-readable instructions, program modules, structured data, or unstructured data.

[0105] Computer-readable storage media can include, but are not limited to, random access

memory (RAM), read only memory (ROM), electrically erasable programmable read only memory (EEPROM), flash memory or other memory technology, solid state drive (SSD) or other solid-state storage technology, compact disk read only memory (CD ROM), digital versatile disk (DVD), Blu-ray disc or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices or other tangible and/or non-transitory media which can be used to store desired information.

[0106] In this regard, the terms “tangible” or “non-transitory” herein as applied to storage, memory or computer-readable media, are to be understood to exclude only propagating transitory signals per se as modifiers and do not relinquish rights to all standard storage, memory or computer-readable media that are not only propagating transitory signals per se. Computer-readable storage media can be accessed by one or more local or remote computing devices, e.g., via access requests, queries or other data retrieval protocols, for a variety of operations with respect to the information stored by the medium.

[0107] On the other hand, communications media typically embody computer-readable instructions, data structures, program modules or other structured or unstructured data in a data signal such as a modulated data signal, e.g., a carrier wave or other transport mechanism, and includes any information delivery or transport media. The term “modulated data signal” or signals refers to a signal that has one or more of its characteristics set or changed in such a manner as to encode information in one or more signals. By way of example, and not limitation, communications media include wired media, such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media

[0108] Further, terms like “user equipment,” “user device,” “mobile device,” “mobile,” station,” “access terminal,” “terminal,” “handset,” and similar terminology, generally refer to a wireless device utilized by a subscriber or user of a wireless communication network or service to receive or convey data, control, voice, video, sound, gaming, or substantially any data-stream or signaling-stream. The foregoing terms are utilized interchangeably in the subject specification and related drawings. Likewise, the terms “access point,” “node B,” “base station,” “evolved Node B,” “cell,” “cell site,” and the like, can be utilized interchangeably in the subject application, and refer to a wireless network component or appliance that serves and receives data, control, voice, video, sound, gaming, or substantially any data-stream or signaling-stream from a set of subscriber stations. Data and signaling streams can be packetized or frame-based flows. It is noted that in the subject specification and drawings, context or explicit distinction provides differentiation with respect to access points or base stations that serve and receive data from a mobile device in an outdoor environment, and access points or base stations that operate in a confined, primarily indoor environment overlaid in an outdoor coverage area. Data and signaling streams can be packetized or frame-based flows.

[0109] Furthermore, the terms “user,” “subscriber,” “customer,” “consumer,” and the like are employed interchangeably throughout the subject specification, unless context warrants particular distinction(s) among the terms. It should be appreciated that such terms can refer to human entities, associated devices, or automated components supported through artificial intelligence (e.g., a capacity to make inference based on complex mathematical formalisms) which can provide simulated vision, sound recognition and so forth. In addition, the terms “wireless network” and “network” are used interchangeable in the subject application, when context wherein the term is utilized warrants distinction for clarity purposes such distinction is made explicit.

[0110] Moreover, the word “exemplary” is used herein to mean serving as an example, instance, or illustration. Any aspect or design described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects or designs. Rather, use of the word exemplary is intended to present concepts in a concrete fashion. As used in this application, the term “or” is intended to mean an inclusive “or” rather than an exclusive “or”. That is, unless specified otherwise, or clear from context, “X employs A or B” is intended to mean any of the

natural inclusive permutations. That is, if X employs A; X employs B; or X employs both A and B, then “X employs A or B” is satisfied under any of the foregoing instances. In addition, the articles “a” and “an” as used in this application and the appended claims should generally be construed to mean “one or more” unless specified otherwise or clear from context to be directed to a singular form.

[0111] In addition, while a particular feature may have been disclosed with respect to only one of several implementations, such feature may be combined with one or more other features of the other implementations as may be desired and advantageous for any given or particular application. Furthermore, to the extent that the terms “includes” and “including” and variants thereof are used in either the detailed description or the claims, these terms are intended to be inclusive in a manner similar to the term “comprising.”

[0112] The above descriptions of various embodiments of the subject disclosure and corresponding figures and what is described in the Abstract, are described herein for illustrative purposes, and are not intended to be exhaustive or to limit the disclosed embodiments to the precise forms disclosed. It is to be understood that one of ordinary skill in the art may recognize that other embodiments having modifications, permutations, combinations, and additions can be implemented for performing the same, similar, alternative, or substitute functions of the disclosed subject matter, and are therefore considered within the scope of this disclosure. Therefore, the disclosed subject matter should not be limited to any single embodiment described herein, but rather should be construed in breadth and scope in accordance with the claims below.

Claims

1. A non-transitory machine-readable medium, comprising executable instructions that, when executed by a processor of network equipment, facilitate performance of operations, the operations comprising: facilitating a configuring of repeater equipment to generate a forwarding layer that is logically above a physical layer of a mobile termination function of the repeater equipment and a distributed unit function of the repeater equipment, wherein the configuring comprises configuring the repeater equipment to route at least data plane signals using the forwarding layer, and wherein the configuring further comprises configuring the repeater equipment to route control plane signals for the mobile termination function and the distributed unit function using the forwarding layer; and facilitating communication of the data plane signals to a user equipment.
2. The non-transitory machine-readable medium of claim 1, wherein the user equipment comprises wireless equipment utilized by a subscriber of a wireless communication service.
3. The non-transitory machine-readable medium of claim 1, wherein the configuring further comprises configuring the repeater equipment to use the forwarding layer to multiplex, in time, signals across channels communicated via at least one of: an access link between the repeater equipment and the user equipment, or a repeater link between the repeater equipment and a parent node.
4. The non-transitory machine-readable medium of claim 1, wherein the configuring further comprises configuring the repeater equipment to use the forwarding layer to multiplex, in frequency, signals across channels communicated via at least one of: an access link between the repeater equipment and the user equipment, or a repeater link between the repeater equipment and a parent node.
5. The non-transitory machine-readable medium of claim 1, wherein the configuring further comprises configuring the repeater equipment to use the forwarding layer to multiplex, spatially, signals across channels communicated via at least one of: an access link between the repeater equipment and the user equipment, or a repeater link between the repeater equipment and a parent node.
6. The non-transitory machine-readable medium of claim 1, wherein the configuring further

comprises configuring the repeater equipment to use the forwarding layer to multiplex, in at least one of: time or frequency or spatially, signals across channels to be used as part of communications by the repeater equipment according to initial access and radio resource management protocol.

7. The non-transitory machine-readable medium of claim 1, wherein the configuring further comprises configuring the repeater equipment to use parameters, of a configuration applicable to synchronization signal blocks of a donor node, to broadcast on an access link between the repeater equipment and the user equipment.

8. The non-transitory machine-readable medium of claim 7, wherein the configuration applicable to synchronization signal blocks of the donor node comprises at least one of: a synchronization signal block timing configuration, or a synchronization signal block measurement timing configuration.

9. The non-transitory machine-readable medium of claim 8, wherein the at least one of the synchronization signal block timing configuration or the synchronization signal block measurement timing configuration indicates, to the repeater equipment, time and frequency resources applicable to the synchronization signal blocks and corresponding indexes for the synchronization signal blocks.

10. The non-transitory machine-readable medium of claim 1, wherein the configuring further comprises configuring the repeater equipment to use a subgroup of parameters, of a group of parameters of a configuration applicable to synchronization signal blocks of a donor node, to broadcast on an access link between the repeater equipment and the user equipment, and wherein the subgroup of parameters indicates, to the repeater equipment, a subgroup of time and frequency resources applicable to the synchronization signal blocks and a corresponding subgroup of indexes for the synchronization signal blocks.

11. The non-transitory machine-readable medium of claim 1, wherein the configuring further comprises configuring the repeater equipment to use at least a subgroup of parameters, of a group of parameters of a configuration, to indicate, to the repeater equipment, at least a subset of time and frequency resources applicable to communications on a random access channel between a donor node and the repeater equipment.

12. The non-transitory machine-readable medium of claim 1, wherein the configuring further comprises configuring the repeater equipment to use a configuration applicable to data signals and control signals and channels utilized for the data signals and the control signals, and wherein the configuration indicates at least one of frequency division duplex bands or time division duplex bands enabled for communication of the data signals and the control signals via the channels.

13. The non-transitory machine-readable medium of claim 1, wherein the configuring the repeater equipment further comprises configuring the repeater equipment to use a configuration, the configuration comprising at least one of: a supported bandwidth part comprising starting frequency location data and size data within a carrier bandwidth on an indication band, an indication of a time division duplex information for downlink signals and uplink signals, a slot pattern comprising a set of consecutive slots, a slot pattern comprising a set of consecutive slots and a set of unavailable slots, signal data and channel data utilized for the signal data, channel state information reference signal data, channel state information reference signal index data, transmission timing data, sounding reference signal resource data, sounding reference signal pattern data, reception timing data, an indication of link adaptation resource data, analog beam data, digital beam data, beamforming weight data for use by the repeater equipment on an access link between the repeater equipment and the user equipment, or beamforming weight data for use by the repeater equipment on a repeater link between the repeater equipment and a parent node.

14. A system, comprising: a processor; and a memory that stores executable instructions that, when executed by the processor of the system, facilitate performance of operations, the operations comprising: in response to receiving by a mobile termination function of repeater equipment, configuration data from a parent node, configuring a forwarding layer logically above a first physical layer of the mobile termination function and a second physical layer of a distributed unit

function of the repeater equipment to forward data plane signals received at the first physical layer via the second physical layer to a user equipment; and managing, by the repeater equipment based on the configuration data, control plane signaling and configuration for the mobile termination function, and control plane signaling and configuration for the distributed unit function.

15. The system of claim 14, wherein the operations further comprise multiplexing, by the repeater equipment based on the configuration data, repeater link data communicated via the first physical layer and access link data communicated via the second physical layer.

16. The system of claim 14, wherein receiving the configuration data comprises at least one of: receiving the configuration data from the parent node based on a schedule, receiving the configuration data from the parent node based on modified parameter data, or receiving the configuration data from the parent node based on a request from the repeater equipment.

17. A non-transitory machine-readable medium, comprising executable instructions that, when executed by a processor, facilitate performance of operations, the operations comprising: receiving, at a repeater link corresponding to a first physical layer of a mobile termination function of repeater equipment, data plane traffic from a parent node, wherein the data plane traffic comprises first data plane traffic; forwarding, via a forwarding layer configured logically above the first physical layer and a second physical layer of a distributed unit function of the repeater equipment, the first data plane traffic to an access link corresponding to the second physical layer; communicating the first data plane traffic, via the access link, to a user equipment; and forwarding, via the forwarding layer, second data plane traffic from the user equipment to the first physical layer for communication of the second data plane traffic to the parent node via the repeater link, wherein the second data plane traffic is received via the access link.

18. The non-transitory machine-readable medium of claim 17, wherein the operations further comprise multiplexing first communications of the repeater link and second communications of the access link.

19. The non-transitory machine-readable medium of claim 17, wherein the user equipment comprises wireless equipment utilized by a subscriber of a wireless communication service.

20. The non-transitory machine-readable medium of claim 17, wherein the processor comprises a single-core processor, a single-processor with software multithread execution capability, a multi-core processor; a multi-core processor with software multithread execution capability, or a multi-core processor with hardware multithread technology.
